

IRRATIONALITY OF AT LEAST ONE OF $\zeta(7), \dots, \zeta(21)$ AND OF AT LEAST ONE OF $\zeta(9), \dots, \zeta(33)$: A COMPLETE PROOF VIA ZUDILIN'S HIGHER-DERIVATIVE CONSTRUCTION

ABSTRACT. We prove: (A) at least one of the eight numbers $\zeta(7), \zeta(9), \dots, \zeta(21)$ is irrational; (B) at least one of the thirteen numbers $\zeta(9), \zeta(11), \dots, \zeta(33)$ is irrational. These sharpen Zudilin's 2002 theorems ($\zeta(7), \dots, \zeta(37)$ and $\zeta(9), \dots, \zeta(53)$). The linear forms are the ones of Zudilin's general construction (*J. Théor. Nombres Bordeaux* 16 (2004), §8) with $r - 1 = 4$, resp. 6, derivatives; the arithmetic of the forms is his Lemma 19, which is valid for every odd r . What was missing for $r \geq 5$ is the analytic lemma (his Lemma 20, proved there only for $r = 3$). We supply it in a form adapted to computer verification: the size of the form is governed by the Fourier mode $e^{\pm i(r-2)\pi t}$ of $\sin^r(\pi t) \cot_r(\pi t)$, whose saddle point is certified together with an explicit steepest-descent path, and a phase condition gives non-vanishing. All finitely many numerical inequalities are listed with the error control used to verify them. The same proof re-derives Zudilin's "one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ " from a slightly different direction vector, as a consistency check.

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1. STATEMENTS

Throughout, $r \geq 3$ and $q \geq r + 4$ are odd integers, and $\boldsymbol{\eta} = (\eta_0; \eta_1, \dots, \eta_q)$ is a vector of positive integers (“directions”) satisfying

$$(1.1) \quad \eta_1 \leq \eta_2 \leq \dots \leq \eta_q < \tfrac{1}{2}\eta_0, \quad \sum_{j=1}^q \eta_j < \tfrac{1}{2}\eta_0 (q - r).$$

For $n \in \mathbb{N}$ put $h_0 = \eta_0 n + 2$, $h_j = \eta_j n + 1$ ($1 \leq j \leq q$) and define the rational function

$$(1.2) \quad R_n(t) = (h_0 + 2t) \prod_{j=1}^r \frac{(t+1)_{h_j-1} (t+h_0-h_j+1)_{h_j-1}}{((h_j-1)!)^2} \prod_{j=r+1}^q \frac{(h_0-2h_j)!}{(t+h_j)_{h_0-2h_j+1}},$$

where $(y)_m = y(y+1) \cdots (y+m-1)$, and the numbers

$$(1.3) \quad F_n = F_n(\boldsymbol{\eta}) = \frac{1}{(r-1)!} \sum_{t=0}^{\infty} R_n^{(r-1)}(t).$$

(This is $F(\mathbf{h})$ of [11, (8.6)–(8.7)]: R_n is the product of Zudilin’s elementary “bricks”, the first r factors are his $\Gamma(h_0+t)\Gamma(h_j+t)/(\Gamma(1+t)\Gamma(1+h_0-h_j+t))$ normalised by $(h_j-1)!^{-2}$.) The first r factors of (1.2) are polynomials in t (the “zero blocks”), the last $q-r$ are the “pole blocks”; the linear form F_n is obtained by differentiating $r-1$ times.

Theorem 1.1 ($r = 5$). *At least one of the eight numbers $\zeta(7), \zeta(9), \zeta(11), \dots, \zeta(21)$ is irrational.*

Theorem 1.2 ($r = 7$). *At least one of the thirteen numbers $\zeta(9), \zeta(11), \zeta(13), \dots, \zeta(33)$ is irrational.*

Theorem 1.3 ($r = 3$; Zudilin [9, 11]). *At least one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational.*

The direction vectors are

$$(A) \quad r = 5, \quad q = 23, \quad \eta_0 = 196, \\ \boldsymbol{\eta}' = (58, 58, 58, 59, 60; 62, 62, 63, 64, 64, 65, 66, 66, 68, 69, 70, 72, 73, 75, 76, 77, 79, 80); \quad (1.4)$$

$$(B) \quad r = 7, \quad q = 35, \quad \eta_0 = 151, \\ \boldsymbol{\eta}' = (46, 46, 46, 46, 46, 46, 46; 47, 47, 48, 49, 49, 49, 50, 51, 51, 51, 52, 53, 53, 53, 54, \\ 55, 55, 55, 56, 57, 57, 57, 58, 59, 59, 59, 60, 61); \quad (1.5)$$

$$(C) \quad r = 3, \quad q = 13, \quad \eta_0 = 90, \quad \boldsymbol{\eta}' = (27, 27, 27; 28, 29, 30, 31, 32, 33, 34, 35, 36, 37); \quad (1.6)$$

where the semicolon separates the r zero-block directions from the $q-r$ pole-block directions. Condition (1.1) holds: $\sum \eta_j = 1544 < 1764$, $1827 < 2114$, $406 < 450$ respectively.

The proof has the usual three parts. (i) F_n is a rational linear form in $1, \zeta(r+2), \zeta(r+4), \dots, \zeta(q-2)$ (Lemma 2.2). (ii) An explicit integer Δ_n with $\Delta_n F_n \in \mathbb{Z} + \mathbb{Z}\zeta(r+2) + \dots + \mathbb{Z}\zeta(q-2)$ and $\lim \frac{1}{n} \log \Delta_n = C_2$ (Section 3; this is Zudilin’s Lemma 19 plus the Chudnovsky–Rukhadze–Hata asymptotics). (iii) $\limsup \frac{1}{n} \log |F_n| \leq -C_0$ and $F_n \neq 0$ infinitely often (Section 4). If all of $\zeta(r+2), \dots, \zeta(q-2)$ were rational with common denominator A , then $A\Delta_n F_n$ would be a non-zero integer for infinitely many n while $|A\Delta_n F_n| \leq e^{n(C_2-C_0+o(1))} \rightarrow 0$ as soon as $C_0 > C_2$. The certified values are

	C_0	C_2	$C_0 - C_2$
(1.7) (A)	939.0301751743	936.8905456295	+2.1396
(B)	1175.7847344814	1171.2347981360	+4.5499
(C)	230.8974422767	227.1012619982	+3.7962

(Section 5; for comparison, Zudilin’s own parameters for Theorem 1.3 give $C_0 = 227.5802$, $C_2 = 226.2494$ [11, p. 285], which our code reproduces to all printed digits.)

2. THE LINEAR FORMS

Lemma 2.1. (a) $R_n(t) = O(t^{-2})$ as $|t| \rightarrow \infty$; more precisely $\deg R_n \leq 1 - q + r - n \leq -2$. (b) $R_n(-t - h_0) = -R_n(t)$. (c) The poles of R_n are the points $t = -k$, $h_{r+1} \leq k \leq h_0 - h_{r+1}$, of order at most $q - r$; R_n has a zero of order exactly r at each $t = -m$, $1 \leq m \leq h_1 - 1$, and at each $t = -m$, $h_0 - h_1 + 1 \leq m \leq h_0 - 1$.

Proof. (a) The numerator has degree $1 + 2 \sum_{j \leq r} (h_j - 1)$ and the denominator has degree $\sum_{j > r} (h_0 - 2h_j + 1)$, so $\deg R_n = 1 - q - r + 2 \sum_{j=1}^q h_j - (q - r)h_0$. With $h_j = \eta_j n + 1$, $h_0 = \eta_0 n + 2$ this is $1 - q + r + 2n(\sum_j \eta_j - \frac{1}{2}(q - r)\eta_0)$. By (1.1) the bracket is a negative half-integer, so $2n(\cdots) \leq -n$ and $\deg R_n \leq 1 - q + r - n \leq -2$ because $q \geq r + 4$. (b) Under $t \mapsto -t - h_0$ the factor $h_0 + 2t$ changes sign; $(t + 1)_{h_j-1} \mapsto (-1)^{h_j-1}(t + h_0 - h_j + 1)_{h_j-1}$ and vice versa, so each zero block is multiplied by $(-1)^{2(h_j-1)} = 1$; each pole block is multiplied by $(-1)^{h_0-2h_j+1} = (-1)^{h_0+1}$. Since $q - r$ is even, the total sign is -1 . (c) The zeros of the j -th zero block are $-1, \dots, -(h_j - 1)$ and $-(h_0 - h_j + 1), \dots, -(h_0 - 1)$; the poles of the j -th pole block are $-h_j, \dots, -(h_0 - h_j)$. Since $h_1 \leq \dots \leq h_q < h_0/2$ these three families are pairwise disjoint from one another in the sense needed: for $1 \leq m \leq h_1 - 1$ all r zero blocks vanish simply at $-m$ and no pole block has a pole there ($h_j \geq h_{r+1} > h_1 - 1$), so the order is exactly r ; similarly at the mirror points. $\square$

By Lemma 2.1(c), $R_n^{(r-1)}(t) = 0$ for $1 - h_1 \leq t \leq -1$, so the sum (1.3) may be started at $t = 1 - h_1$; this is the form in which the arithmetic is analysed. Write the partial fraction expansion

$$(2.1) \quad R_n(t) = \sum_{i=1}^{q-r} \sum_{k=h_{r+1}}^{h_0-h_{r+1}} \frac{a_{i,k}}{(t+k)^i}, \quad a_{i,k} = \frac{1}{(q-r-i)!} \left(\frac{d}{dt} \right)^{q-r-i} \left[R_n(t)(t+k)^{q-r} \right]_{t=-k}.$$

Lemma 2.2. $a_{i,k} = (-1)^{i+1} a_{i,h_0-k}$; $\sum_k a_{1,k} = 0$; and

$$(2.2) \quad \begin{aligned} F_n &= \sum_{\substack{r+2 \leq s \leq q-2 \\ s \text{ odd}}} A_{s,n} \zeta(s) - A_{0,n}, \quad A_{s,n} = \binom{s-1}{r-1} \sum_k a_{s-r+1,k}, \\ A_{0,n} &= \sum_{i=1}^{q-r} \binom{i+r-2}{r-1} \sum_k a_{i,k} \sum_{\ell=1}^{k-h_1} \frac{1}{\ell^{i+r-1}}. \end{aligned}$$

In particular $F_n \in \mathbb{Q} + \mathbb{Q}\zeta(r+2) + \mathbb{Q}\zeta(r+4) + \dots + \mathbb{Q}\zeta(q-2)$.

Proof. Differentiating (2.1) $r-1$ times and dividing by $(r-1)!$ gives

$$\frac{R_n^{(r-1)}(t)}{(r-1)!} = (-1)^{r-1} \sum_{i,k} \binom{i+r-2}{r-1} \frac{a_{i,k}}{(t+k)^{i+r-1}},$$

and $(-1)^{r-1} = 1$. Summing over $t \geq 1 - h_1$ and using $\sum_{t \geq 1-h_1} (t+k)^{-m} = \zeta(m) - \sum_{\ell=1}^{k-h_1} \ell^{-m}$ for $m \geq 2$ (note $k \geq h_{r+1} > h_1$, so $t+k \geq 1$), while for $m = 1$ (i.e. $i = 1$, $r = 1$ — which does not occur since $r \geq 3$) nothing is needed, we get (2.2) with all i from 1 to $q-r$. The symmetry $a_{i,k} = (-1)^{i+1} a_{i,h_0-k}$ follows from Lemma 2.1(b) and uniqueness of (2.1); hence $\sum_k a_{i,k} = 0$ for even i , i.e. $A_{s,n} = 0$ for even $s = i + r - 1$. Finally $\sum_k a_{1,k}$ is the sum of all residues of R_n , which vanishes because $R_n = O(t^{-2})$; this kills $s = r$. The surviving odd $s = i + r - 1$ with $3 \leq i \leq q - r$ odd are $r+2, r+4, \dots, q-2$. $\square$

3. ARITHMETIC OF THE COEFFICIENTS

Let $D_N = \text{lcm}(1, 2, \dots, N)$. Put

$$(3.1) \quad \hat{m}_0 = \max\{\eta_r, \eta_0 - 2\eta_{r+1}\}, \quad \hat{m}_j = \max\{\hat{m}_0, \eta_0 - \eta_1 - \eta_{r+j}\} \quad (1 \leq j \leq q-r),$$

so that $\hat{m}_1 \geq \hat{m}_2 \geq \dots \geq \hat{m}_{q-r} \geq \hat{m}_0$. (In Zudilin's notation $m_0 = \max\{h_r - 1, h_0 - 2h_{r+1}\} = \hat{m}_0 n$ and $m_j = \max\{m_0, h_0 - h_1 - h_{r+j}\} = \hat{m}_j n$ exactly, because $h_0 - h_1 - h_{r+j} = (\eta_0 - \eta_1 - \eta_{r+j})n$.)

3.1. The function φ . For real x, y let

$$(3.2) \quad \begin{aligned} \varphi_0(x, y) = & \sum_{j=1}^r \left(\lfloor y \rfloor + \lfloor \eta_0 x - y \rfloor - \lfloor y - \eta_j x \rfloor - \lfloor (\eta_0 - \eta_j)x - y \rfloor - 2\lfloor \eta_j x \rfloor \right) \\ & + \sum_{j=r+1}^q \left(\lfloor (\eta_0 - 2\eta_j)x \rfloor - \lfloor y - \eta_j x \rfloor - \lfloor (\eta_0 - \eta_j)x - y \rfloor \right), \end{aligned}$$

and $\varphi(x) = \min_{y \in \mathbb{R}} \varphi_0(x, y)$.

Lemma 3.1. φ_0 is 1-periodic in each variable, integer valued and ≥ 0 ; hence φ is 1-periodic, integer valued, ≥ 0 , and the minimum is attained. Moreover φ is constant on each open interval between consecutive points of the finite set

$$X = \{a/b \in [0, 1] : a \in \mathbb{Z}, b \in Q\}, \quad Q = \{|a_i b_{i'} - a_{i'} b_i| \neq 0\} \cup \{|a_i| : b_i = 0\} \cup \{\hat{m}_{q-r}\},$$

where (a_i, b_i) runs over the coefficient pairs of the linear forms $a_i x + b_i y$ appearing in (3.2), i.e. $(0, 1), (\eta_0, -1), (-\eta_j, 1), (\eta_0 - \eta_j, -1), (\eta_j, 0)$ for $j \leq r$ and $(\eta_0 - 2\eta_j, 0), (-\eta_j, 1), (\eta_0 - \eta_j, -1)$ for $j > r$.

Proof. Periodicity in y : the coefficients of y in the four y -dependent terms of each zero block sum to $1 - 1 - 1 + 1 = 0$ and in each pole block to $-1 + 1 = 0$; in x : $\eta_0 - (-\eta_j) \cdot 1 \dots$, explicitly each zero block contributes $\eta_0 + \eta_j - (\eta_0 - \eta_j) - 2\eta_j = 0$ and each pole block $(\eta_0 - 2\eta_j) + \eta_j - (\eta_0 - \eta_j) = 0$. Non-negativity: each pole-block summand is $\lfloor A + B \rfloor - \lfloor A \rfloor - \lfloor B \rfloor \in \{0, 1\}$ with $A = y - \eta_j x$, $B = (\eta_0 - \eta_j)x - y$; each zero-block summand is the sum of two such expressions, with $(A, B) = (y - \eta_j x, \eta_j x)$ and $((\eta_0 - \eta_j)x - y, \eta_j x)$. For fixed x the function $y \mapsto \varphi_0(x, y)$ is piecewise constant with jumps only on the lines $a_i x + b_i y \in \mathbb{Z}$, $b_i \neq 0$; as x varies, the value of each $\lfloor a_i x + b_i y \rfloor$ on a cell of the arrangement of lines $\{a_i x + b_i y = k\}$ is constant, so the collection of values of $\varphi_0(x, \cdot)$ can only change at abscissae where two lines meet, $x = (kb_{i'} - k'b_i)/(a_i b_{i'} - a_{i'} b_i)$, or where a vertical line $a_i x = k$ ($b_i = 0$) is crossed. These abscissae lie in X . $\square$

For a prime p and $n \geq 1$ put $\Phi_n = \prod_{\sqrt{h_0} < p \leq \hat{m}_{q-r} n} p^{\varphi(n/p)}$ and

$$(3.3) \quad \Delta_n = D_{\hat{m}_1 n}^r \prod_{j=2}^{q-r} D_{\hat{m}_j n} \cdot \Phi_n^{-1}.$$

Lemma 3.2 (Zudilin [11, Lemma 19]). *For all $n \geq n_0(\eta)$, Δ_n is a positive integer and $\Delta_n A_{s,n} \in \mathbb{Z}$ for $s = 0$ and for all odd s with $r + 2 \leq s \leq q - 2$. Consequently $\Delta_n F_n \in \mathbb{Z} + \mathbb{Z}\zeta(r + 2) + \mathbb{Z}\zeta(r + 4) + \dots + \mathbb{Z}\zeta(q - 2)$.*

Proof. This is exactly [11, Lemma 19] (whose proof is written for arbitrary odd r , $q \geq r + 4$ and directions satisfying (1.1)), with the minor difference that Zudilin's Φ has exponents $\nu_p = \min_{h_{r+1} \leq k \leq h_0 - h_{r+1}} \nu_{k,p}$ where $\nu_{k,p}$ is a sum of floor functions of the quantities $(k-1)/p$, $(h_0 - k - 1)/p$, $(k - h_j)/p$, $(h_0 - h_j - k)/p$, $(h_j - 1)/p$, $(h_0 - 2h_j)/p$; with $h_0 = \eta_0 n + 2$, $h_j = \eta_j n + 1$ one checks term by term that $\nu_{k,p} = \varphi_0(n/p, (k-1)/p)$, so $\nu_p \geq \varphi(n/p) \geq 0$ and $\Phi_n \mid \Phi$; dividing by the smaller integer Φ_n instead of Φ preserves the conclusion. For the reader's convenience we recall the argument. Write $R_n(t)(t+k)^{q-r}$ as the product of $h_0 + 2t$, the $2r$ integer-valued polynomial bricks $P_j^\pm(t) = (t+1)_{h_j-1}/(h_j-1)!$, $(t+h_0-h_j+1)_{h_j-1}/(h_j-1)!$ and the $q-r$ factors $C_j(t)(t+k)$. By the Leibniz rule, $a_{i,k}$ in (2.1) is a sum of products $\prod_\beta \frac{1}{l_\beta!} (\text{brick}_\beta)^{(l_\beta)} (-k)$ over all distributions $\sum_\beta l_\beta = q - r - i$ of the derivatives among the bricks. By [11, Lemmas

15, 16], $D_{\hat{m}_0 n}^l \frac{1}{l!} (P_j^\pm)^{(l)}(-k) \in \mathbb{Z}$ and $D_{\hat{m}_0 n}^l \frac{1}{l!} (C_j(t)(t+k))^{(l)}|_{t=-k} \in \mathbb{Z}$ for every $l \geq 0$ and every $k \in [h_{r+1}, h_0 - h_{r+1}]$ (the lengths of the bricks are $h_j - 1 \leq \hat{m}_0 n$ and $h_0 - 2h_j \leq \hat{m}_0 n$), whence

$$(3.4) \quad D_{\hat{m}_0 n}^{q-r-i} a_{i,k} \in \mathbb{Z}.$$

By [11, Lemmas 17, 18] (applied with $p > \sqrt{h_0}$, so that $\text{ord}_p N! = \lfloor N/p \rfloor$ for all $N \leq h_0$ and $p \nmid l!$ for $l \leq q$ once $n \geq n_0$), the p -adic order of each brick derivative $\frac{1}{l!}(\text{brick})^{(l)}(-k)$ is at least $-l$ plus the corresponding summand of $\varphi_0(n/p, (k-1)/p)$; summing over the bricks and taking the minimum over the Leibniz terms,

$$(3.5) \quad \text{ord}_p a_{i,k} \geq -(q-r-i) + \varphi_0(n/p, (k-1)/p) \geq -(q-r-i) + \varphi(n/p) \quad (p > \sqrt{h_0}).$$

The coefficients $A_{s,n}$, $s \geq r+2$. Here $A_{s,n} = \binom{s-1}{r-1} \sum_k a_{i,k}$ with $i = s - r + 1 \geq 3$, so by (3.4) already $D_{\hat{m}_0 n}^{q-r-i} A_{s,n} \in \mathbb{Z}$, and $D_{\hat{m}_0 n}^{q-r-i}$ divides the product of D 's in (3.3).

The constant term $A_{0,n}$. Consider the term $a_{i,k} \sum_{\ell \leq k-h_1} \ell^{-(i+r-1)}$ of (2.2). If $a_{i,k} \neq 0$ then $-k$ is a pole of R_n of order $\geq i$, i.e. k lies in the ranges $[h_j, h_0 - h_j]$ of at least i pole blocks; these ranges are nested and decreasing in j , hence $k \leq h_0 - h_{r+i}$ and $k - h_1 \leq h_0 - h_1 - h_{r+i} = (\eta_0 - \eta_1 - \eta_{r+i})n \leq \hat{m}_i n$. Thus the term has denominator dividing $D_{\hat{m}_i n}^{i+r-1} D_{\hat{m}_0 n}^{q-r-i}$. The product $D_{\hat{m}_1 n}^r D_{\hat{m}_2 n} \cdots D_{\hat{m}_{q-r} n}$ has $q-1$ factors; its $i+r-1$ largest factors $D_{\hat{m}_1 n}^r D_{\hat{m}_2 n} \cdots D_{\hat{m}_i n}$ are all divisible by $D_{\hat{m}_i n}$ and the remaining $q-r-i$ factors are divisible by $D_{\hat{m}_0 n}$. Hence the product of D 's in (3.3) times $A_{0,n}$ is an integer.

Removing Φ_n . Let $\sqrt{h_0} < p \leq \hat{m}_{q-r} n$. Since $\hat{m}_1 n < \eta_0 n < h_0 < p^2$, each of the $q-1$ factors $D_{\hat{m}_j n}$ is divisible by p exactly once, and every $\ell \leq \hat{m}_1 n$ has $\text{ord}_p \ell \leq 1$. By (3.5), $\text{ord}_p A_{s,n} \geq -(q-r-i) + \varphi(n/p)$ and $\text{ord}_p A_{0,n} \geq \min_i (-(q-r-i) + \varphi(n/p) - (i+r-1)) = -(q-1) + \varphi(n/p)$. The p -part of Δ_n is $p^{q-1-\varphi(n/p)}$, so $\text{ord}_p(\Delta_n A_{s,n}) \geq 0$ for all s . For the other primes the p -part of Δ_n equals that of the product of the D 's, and the previous two paragraphs apply. Finally $\Delta_n \in \mathbb{Z}$ because $\varphi \leq q-1$ (checked in §3.3: $\max \varphi = 18, 27, 11$ against $q-1 = 22, 34, 12$). $\square$

3.2. Asymptotics of Δ_n .

Lemma 3.3. $\lim_{n \rightarrow \infty} \frac{\log \Delta_n}{n} = C_2 := r\hat{m}_1 + \hat{m}_2 + \cdots + \hat{m}_{q-r} - I_\Phi$, where

$$(3.6) \quad I_\Phi = \int_0^1 \varphi(x) \psi'(1+x) dx + \int_{1/\hat{m}_{q-r}}^1 \frac{\varphi(x)}{x^2} dx = \int_0^1 \varphi(x) d\psi(x) - \int_0^{1/\hat{m}_{q-r}} \frac{\varphi(x)}{x^2} dx,$$

$\psi = \Gamma'/\Gamma$. If $0 = x_0 < x_1 < \cdots < x_N = 1$ are the points of X and $\varphi = \varphi_i$ on (x_{i-1}, x_i) , then

$$(3.7) \quad I_\Phi = \sum_{i=1}^N \varphi_i(\psi(1+x_i) - \psi(1+x_{i-1})) + \sum_{i: x_{i-1} \geq 1/\hat{m}_{q-r}} \varphi_i \left(\frac{1}{x_{i-1}} - \frac{1}{x_i} \right).$$

Proof. $\log D_N = N(1+o(1))$ is the prime number theorem. For Φ_n we use the lemma of Chudnovsky–Rukhadze–Hata in the form of [10, Lemma 4.4] (see also [4, Lemma 3.2]): for $0 < u < v < 1$ and $c > 0$, $\sum_{p > \sqrt{cn}, \{n/p\} \in [u,v]} \log p = n(\psi(v) - \psi(u)) + o(n)$. The primes $p > n$ with $n/p \in [u, v]$ contribute $n(1/u - 1/v) + o(n)$ by the prime number theorem, so the primes $p \leq n$ with $\{n/p\} \in [u, v]$ contribute $n \int_u^v \psi'(1+x) dx + o(n)$ (recall $\psi'(x) = \psi'(1+x) + x^{-2}$), and the primes $n < p \leq \hat{m}_{q-r} n$ with $n/p \in [u, v]$ contribute $n \int_{[u,v] \cap [1/\hat{m}_{q-r}, 1]} x^{-2} dx + o(n)$. Apply this to the finitely many intervals (x_{i-1}, x_i) on which φ is constant (Lemma 3.1; $1/\hat{m}_{q-r} \in X$); the primes with $n/p \in X$ are at most $|X|$ in number for each n and contribute $O(\log n)$. Since $\varphi \geq 0$ is bounded, the excluded primes $p \leq \sqrt{h_0}$ contribute $O(\sqrt{n} \log n)$. Formula (3.7) is (3.6) evaluated on the partition. $\square$

3.3. Numerical values. For the three direction vectors (1.4)–(1.6) the quantities in Lemma 3.3 are as follows ($\hat{m} = (\hat{m}_1, \dots, \hat{m}_{q-r})$; $|X|$ is the number of breakpoints; $\max \varphi$ the largest value of φ on $[0, 1)$):

	$\hat{m}$	$r\hat{m}_1 + \sum_{j \geq 2} \hat{m}_j$	$ X $	$\max \varphi$	I_Φ	C_2
(A)	(76, 76, 75, 74, 74, 73, $72^{\times 12}$)	1616	3461	18	679.1094543705	936.8905456295
(B)	(58, 58, $57^{\times 26}$)	1946	2231	27	774.7652018640	1171.2347981360
(C)	(35, $34^{\times 9}$)	411	879	11	183.8987380018	227.1012619982

How these were computed and how accurate they are. The set X was generated exactly (all denominators in Q are $\leq 2\eta_0 + \hat{m}_{q-r}$). On each interval, $\varphi_i = \min_y \varphi_0(\hat{x}, y)$ at the midpoint $\hat{x}$ was computed twice: once in floating point and once in exact rational arithmetic (the minimum over y is a minimum over the finitely many cells of $y \mapsto \varphi_0(\hat{x}, y)$, whose endpoints are the fractional parts of $-a_i \hat{x}/b_i$); the two computations agree on every interval. Thus the φ_i are exact integers. The values $\psi(1 + x_i)$ were evaluated to 30 significant digits (mpmath); the sums (3.7) were then accumulated in double precision. With at most 3461 terms, each of absolute value $\leq 27 \cdot \max |\psi'| \leq 27 \cdot \pi^2/6$, the accumulated rounding error is below 10^{-9} , while the margins $C_0 - C_2$ in (1.7) exceed 2. (Zudilin's parameters for Theorem 1.3, $\eta_0 = 91$, $\eta_j = 27, 27, 27, 29, 30, \dots, 38$, give with the same code $I_\Phi = 176.75055734$ and $C_2 = 226.24944266$, in agreement with [11, p. 285] to all published digits.)

4. ANALYTIC ESTIMATES

4.1. Notation. We scale $t = nu$. Let $\Omega = \mathbb{C} \setminus ((-\infty, -\eta_1] \cup [0, \infty))$ and, with principal branches of the logarithm,

$$(4.1) \quad f(u) = r \left[(-u) \log(-u) + (\eta_0 + u) \log(\eta_0 + u) \right] + \sum_{j=1}^q \left[(\eta_j + u) \log(\eta_j + u) - (\eta_0 - \eta_j + u) \log(\eta_0 - \eta_j + u) \right] + \kappa_0,$$

$\kappa_0 = \sum_{j > r} (\eta_0 - 2\eta_j) \log(\eta_0 - 2\eta_j) - 2 \sum_{j \leq r} \eta_j \log \eta_j$. Then f is analytic on Ω , real on $(-\eta_1, 0)$, $f(\bar{u}) = \overline{f(u)}$, and

$$(4.2) \quad f'(u) = r \left[\log(\eta_0 + u) - \log(-u) \right] + \sum_j \left[\log(\eta_j + u) - \log(\eta_0 - \eta_j + u) \right],$$

$$(4.3) \quad f''(u) = r \left[\frac{1}{\eta_0 + u} - \frac{1}{u} \right] + \sum_j \left[\frac{1}{\eta_j + u} - \frac{1}{\eta_0 - \eta_j + u} \right],$$

$$f'''(u) = -r \left[\frac{1}{(\eta_0 + u)^2} - \frac{1}{u^2} \right] - \sum_j \left[\frac{1}{(\eta_j + u)^2} - \frac{1}{(\eta_0 - \eta_j + u)^2} \right].$$

Put $G(u) = (\eta_0 + 2u)(-u)^{-r/2}(\eta_0 + u)^{3r/2} \prod_{j=1}^q (\eta_j + u)^{1/2} (\eta_0 - \eta_j + u)^{-3/2}$ (principal powers; analytic and non-vanishing on $\Omega \setminus \{-\eta_0/2\}$), and for $k \in \mathbb{Z}$

$$F_k(u) = f(u) + ik\pi u, \quad H_k(u) = \operatorname{Re} F_k(u) = \operatorname{Re} f(u) - k\pi \operatorname{Im} u.$$

4.2. Integral representation and Fourier modes. Let $\cot_r(z) = \frac{(-1)^{r-1}}{(r-1)!} \frac{d^{r-1}}{dz^{r-1}} \cot z$. By [10, Lemmas 2.2, 2.3], $\cot_r z = U_r(\cot z)$ for a polynomial U_r of degree r , and $\pi^r \cot_r(\pi t) = (t - m)^{-r} + O(1)$ near every $m \in \mathbb{Z}$; in particular $\cot_r(\pi t)$ is bounded on $\{|\operatorname{Im} t| \geq 1\}$ and on the lines $\operatorname{Re} t \in \frac{1}{2} + \mathbb{Z}$.

Lemma 4.1. *Let $M \in \mathbb{R}$ with $-h_1 < M < 0$. Then*

$$(4.4) \quad F_n = -\frac{1}{2\pi i} \int_{M-i\infty}^{M+i\infty} \pi^r \cot_r(\pi t) R_n(t) dt.$$

Proof. On the closed half plane $\operatorname{Re} t \geq M$ the function $\pi^r \cot_r(\pi t) R_n(t)$ is meromorphic with poles exactly at $t = 0, 1, 2, \dots$ (at the integers $-1, \dots, -(h_1 - 1)$ the pole of $\cot_r$ of order r is cancelled by the zero of R_n of order r , Lemma 2.1(c); the poles of R_n lie in $\operatorname{Re} t \leq -h_{r+1} < M$), and $\operatorname{Res}_{t=m} = R_n^{(r-1)}(m)/(r-1)!$ by the expansion of $\pi^r \cot_r(\pi t)$ at m . Integrate over the boundary of the rectangle with vertices $M \pm iN$, $N' + \frac{1}{2} \pm iN$: on the three sides other than $\operatorname{Re} t = M$ the factor $\cot_r(\pi t)$ is bounded and $R_n(t) = O(|t|^{-2})$, so these sides contribute $O(N^{-1} + N'^{-1})$. Letting $N, N' \rightarrow \infty$ and keeping track of the orientation (the side $\operatorname{Re} t = M$ is traversed downwards) gives (4.4). This is [10, Lemma 2.4] in our normalisation. $\square$

Lemma 4.2 (Eulerian expansion). *For odd $r \geq 3$,*

$$(4.5) \quad \sin^r(z) \cot_r(z) = \sum_{\substack{|k| \leq r-2 \\ k \equiv r \pmod{2}}} c_k e^{ikz}, \quad c_k = c_{-k} = \frac{A(r-1, \frac{r-2-k}{2})}{(r-1)!} > 0,$$

where $A(m, j)$ are the Eulerian numbers; in particular $c_{r-2} = 1/(r-1)!$. Explicitly $\sin^3 z \cot_3 z = \cos z$, $\sin^5 z \cot_5 z = \frac{1}{24}(e^{3iz} + 11e^{iz} + 11e^{-iz} + e^{-3iz})$, $\sin^7 z \cot_7 z = \frac{1}{720}(e^{5iz} + 57e^{3iz} + 302e^{iz} + 302e^{-iz} + 57e^{-3iz} + e^{-5iz})$.

Proof. Write $x = e^{2iz}$, so $\cot z = i + \frac{2i}{x-1}$ and $\frac{d}{dz} = 2ix \frac{d}{dx}$. The Eulerian identity $\sum_{\nu \geq 0} \nu^m x^\nu = \frac{x A_m(x)}{(1-x)^{m+1}}$, $A_m(x) = \sum_{j=0}^{m-1} A(m, j) x^j$ [3, (6.37)], gives $(x \frac{d}{dx})^m \frac{1}{1-x} = -\frac{x A_m(x)}{(1-x)^{m+1}}$ and hence $\cot^{(m)}(z) = -2i(2i)^m \frac{x A_m(x)}{(1-x)^{m+1}}$ for $m \geq 1$. With $m = r-1$ and $\sin^r z = e^{-irz}(-1)^r(1-x)^r/(2i)^r$ one obtains $\sin^r z \cot_r z = \frac{1}{(r-1)!} e^{-irz} x A_{r-1}(x) = \frac{1}{(r-1)!} \sum_{j=0}^{r-2} A(r-1, j) e^{i(2j+2-r)z}$. The symmetry $A(m, j) = A(m, m-1-j)$ gives $c_k = c_{-k}$; $A(m, 0) = 1$ gives $c_{\pm(r-2)} = 1/(r-1)!$. The explicit cases are $A(2, \cdot) = (1, 1)$, $A(4, \cdot) = (1, 11, 11, 1)$, $A(6, \cdot) = (1, 57, 302, 302, 57, 1)$. $\square$

Next we write R_n through Gamma functions. Since $(t+1)_{h_j-1} = \Gamma(t+h_j)/\Gamma(t+1)$, $(t+h_0-h_j+1)_{h_j-1} = \Gamma(t+h_0)/\Gamma(t+h_0-h_j+1)$, $(t+h_j)_{h_0-2h_j+1} = \Gamma(t+h_0-h_j+1)/\Gamma(t+h_j)$ and $1/\Gamma(1+t) = -\Gamma(-t) \sin(\pi t)/\pi$,

$$(4.6) \quad R_n(t) = \left(-\frac{\sin \pi t}{\pi} \right)^r \mathcal{G}(t), \quad \mathcal{G}(t) := \operatorname{Norm} \cdot (h_0 + 2t) \Gamma(-t)^r \Gamma(t+h_0)^r \prod_{j=1}^q \frac{\Gamma(t+h_j)}{\Gamma(t+h_0-h_j+1)},$$

$$\operatorname{Norm} = \frac{\prod_{j>r}(h_0-2h_j)!}{\prod_{j \leq r}(h_j-1)!^2}.$$

$\mathcal{G}$ is meromorphic; in the half plane $\operatorname{Re} t > -h_1$ its only poles are $t = 0, 1, 2, \dots$ (from $\Gamma(-t)^r$), and $\mathcal{G}(\bar{t}) = \overline{\mathcal{G}(t)}$. As r is odd, $\pi^r \cot_r(\pi t) R_n(t) = -\sin^r(\pi t) \cot_r(\pi t) \mathcal{G}(t)$, and Lemmas 4.1, 4.2 give

$$(4.7) \quad F_n = \frac{1}{2\pi i} \sum_k c_k I_{n,k}, \quad I_{n,k} := \int_{M-i\infty}^{M+i\infty} e^{ik\pi t} \mathcal{G}(t) dt, \quad \frac{I_{n,-k}}{2\pi i} = \overline{\left(\frac{I_{n,k}}{2\pi i} \right)},$$

the last identity because $\mathcal{G}(\bar{t}) = \overline{\mathcal{G}(t)}$ and the line is mapped to itself with reversed orientation by conjugation. Hence

$$(4.8) \quad F_n = 2 \sum_{k>0, k \equiv r \pmod{2}}^{r-2} c_k \operatorname{Re} \left[\frac{1}{2\pi i} I_{n,k} \right].$$

4.3. Stirling's formula on admissible sets.

Lemma 4.3. *Let $K \subset \Omega$ be a set with $\inf_{u \in K} |u| > 0$, $\inf_{u \in K} \operatorname{Re} u > -\eta_1$ and $\sup_{u \in K} |\arg(-u)| < \pi$. Then there are C_K and, for each $n \geq 1$, a function ε_n on K with $|\varepsilon_n(u)| \leq C_K/n$, such that*

$$(4.9) \quad \mathcal{G}(nu) = \mathcal{A}_n e^{nf(u)} G(u) (1 + \varepsilon_n(u)), \quad \mathcal{A}_n = c_0 n^{1-r-(q-r)/2}, \quad c_0 = (2\pi)^{\frac{q-r}{2}} \frac{\prod_{j>r} (\eta_0 - 2\eta_j)^{1/2}}{\prod_{j \leq r} \eta_j}.$$

Proof. With $t = nu$ the arguments of the Gamma functions in (4.6) are $-t = n(-u)$, $t + h_0 = n(\eta_0 + u) + 2$, $t + h_j = n(\eta_j + u) + 1$, $t + h_0 - h_j + 1 = n(\eta_0 - \eta_j + u) + 2$, and $\operatorname{Norm} = \prod_{j>r} \Gamma(n(\eta_0 - 2\eta_j) + 1) / \prod_{j \leq r} \Gamma(n\eta_j + 1)^2$. Using $\Gamma(z+1) = z\Gamma(z)$ we reduce all of them to $\Gamma(n\zeta)$ with $\zeta \in \{-u, \eta_0 + u, \eta_j + u, \eta_0 - \eta_j + u, \eta_0 - 2\eta_j, \eta_j\}$, at the price of the explicit polynomial factors $(v+1)v$, w , $(w'+1)w'$, etc. On K every such ζ satisfies $|\zeta| \geq c > 0$ and $|\arg \zeta| \leq \pi - \varepsilon_0$ (for $\zeta = -u$ by hypothesis; the others have positive real part because $\operatorname{Re} u > -\eta_1$). By Stirling's formula in the form $\Gamma(z) = \sqrt{2\pi} z^{z-1/2} e^{-z} (1 + \rho(z))$ with $|\rho(z)| \leq C_{\varepsilon_0}/|z|$ uniformly in $|\arg z| \leq \pi - \varepsilon_0$ [8, §12.33], [6, Ch. 8, §4], applied with $z = n\zeta$ and principal logarithms ($\log(n\zeta) = \log n + \log \zeta$), the exponent $n \sum \pm \zeta \log \zeta$ is exactly $nf(u)$ (the sum of the coefficients of ζ over all Gamma factors, counted with multiplicity, is 0, so the terms $n\zeta \log n$ and $-n\zeta$ cancel and κ_0 is contributed by Norm). Collecting the factors $\sqrt{2\pi}$, the powers $(n\zeta)^{-1/2}$ and the polynomial factors gives $\mathcal{A}_n$, $G(u)$ and a factor $(1 + O(1/n))$; a direct count yields the exponent $1 - r - (q - r)/2$ of n and the constant c_0 . $\square$

Lemma 4.4 (decay). *Fix n and $\varrho > 0$. There are C, c (depending on n, ϱ) such that for $|\operatorname{Re} t| \leq \varrho n$, $|\operatorname{Im} t| \geq 1$ and $|k| \leq r - 2$: $|e^{ik\pi t} \mathcal{G}(t)| \leq C(1 + |t|)^c e^{-2\pi |\operatorname{Im} t|}$.*

Proof. For x in a compact set and $|y| \geq 1$, $|\Gamma(x + iy)| \leq C|y|^{x-1/2} e^{-\pi|y|/2}$ [8, §12.33]. In (4.6) the factors $\Gamma(-t)^r \Gamma(t + h_0)^r$ contribute $e^{-r\pi |\operatorname{Im} t|}$, the ratios $\Gamma(t + h_j)/\Gamma(t + h_0 - h_j + 1)$ contribute e^0 , and $|e^{ik\pi t}| = e^{-k\pi \operatorname{Im} t} \leq e^{(r-2)\pi |\operatorname{Im} t|}$. $\square$

4.4. A Laplace lemma.

Lemma 4.5. *Let $\mathcal{L}$ be a piecewise smooth path, F and G analytic on a neighbourhood of $\mathcal{L}$, and let $u_* \in \mathcal{L}$, $\phi \in \mathbb{R}$, $s_0, \delta > 0$ be such that*

- (i) $F'(u_*) = 0$, $F''(u_*) \neq 0$, $e^{2i\phi} F'''(u_*) = -|F''(u_*)|$, $G(u_*) \neq 0$, and the segment $\sigma = \{u_* + se^{i\phi} : |s| \leq s_0\}$ is contained in $\mathcal{L}$ (traversed in the direction of increasing s);
- (ii) $\operatorname{Re} F(u_* + se^{i\phi}) \leq \operatorname{Re} F(u_*) - \frac{1}{4}|F''(u_*)|s^2$ for $|s| \leq s_0$;
- (iii) $\operatorname{Re} F(u) \leq \operatorname{Re} F(u_*) - \delta$ for $u \in \mathcal{L} \setminus \sigma$;
- (iv) $J := \int_{\mathcal{L}} |e^{F(u)} G(u)| |du| < \infty$.

Let ε_n be measurable on $\mathcal{L}$ with $|\varepsilon_n| \leq C/n$. Then, as $n \rightarrow \infty$,

$$\int_{\mathcal{L}} e^{nF(u)} G(u) (1 + \varepsilon_n(u)) du = e^{nF(u_*)} \left[e^{i\phi} G(u_*) \sqrt{\frac{2\pi}{n|F''(u_*)|}} + o(n^{-1/2}) \right].$$

Proof. On $\mathcal{L} \setminus \sigma$ we have $|e^{n(F-F(u_*))}| \leq e^{-(n-1)\delta} |e^{F-F(u_*)}|$ by (iii), so that part of the integral is $O(e^{-n\delta} J)$ relative to $e^{nF(u_*)}$. On σ put $u = u_* + se^{i\phi}$; then $F(u) - F(u_*) = -\frac{1}{2}|F''(u_*)|s^2 + \rho(s)$ with $|\rho(s)| \leq M_3|s|^3/6$, $M_3 = \max_{|w| \leq s_0} |F'''(u_* + w)|$, and $G(u) = G(u_*) + O(|s|)$. Let $s_n = n^{-2/5}$. For $s_n \leq |s| \leq s_0$, (ii) bounds the integrand by $e^{-n|F''(u_*)|s_n^2/4} \max_{\sigma} |G|(1 + C/n)$, which is $O(e^{-cn^{1/5}})$. For $|s| \leq s_n$: $n|\rho(s)| \leq M_3 n^{-1/5}/6$, so $e^{n\rho} = 1 + O(n^{-1/5})$, $G = G(u_*)(1 + O(n^{-2/5}))$, $1 + \varepsilon_n = 1 + O(n^{-1})$, and $\int_{-s_n}^{s_n} e^{-n|F''(u_*)|s^2/2} ds = \sqrt{2\pi/(n|F''(u_*)|)} (1 + O(e^{-cn^{1/5}}))$. Multiplying by $du = e^{i\phi} ds$ gives the claim. $\square$

4.5. Deformation of the contour.

Lemma 4.6. *Let $x_0 \in (-\eta_1, 0)$ and let $\mathcal{L}$ be a piecewise smooth path from $-i\infty$ to $+i\infty$ which meets the real axis only at x_0 , satisfies $\inf_{\mathcal{L}} \operatorname{Re} u > -\eta_1$, and coincides with vertical rays $\{\operatorname{Re} u = \text{const}\}$ outside a compact set. Then for every n and $|k| \leq r-2$, with $M = nx_0$ in (4.7),*

$$I_{n,k} = n \int_{\mathcal{L}} e^{ik\pi nu} \mathcal{G}(nu) du.$$

Proof. The integrand $u \mapsto e^{ik\pi nu} \mathcal{G}(nu)$ is meromorphic on $\{\operatorname{Re} u > -\eta_1\}$ with poles only at $u \in \frac{1}{n}\mathbb{Z}_{\geq 0} \subset [0, \infty)$. The vertical line $\operatorname{Re} u = x_0$ and $\mathcal{L}$ both cross the real axis exactly once, at $x_0 < 0$; the region between them (in each half plane) is therefore contained in $\{\operatorname{Im} u \neq 0\} \cup \{x_0\}$ and contains no pole. Truncate at $|\operatorname{Im} u| = Y'$ and close with horizontal segments; by Lemma 4.4 these segments contribute $O(Y'^c e^{-2\pi n Y'}) \rightarrow 0$. Cauchy's theorem gives the claim. $\square$

4.6. The main analytic proposition.

Proposition 4.7. *Let $k_d = r-2$, $F = F_{k_d}$. Suppose there exist $x_0 \in (-\eta_1, 0)$, a path $\mathcal{L}$ as in Lemma 4.6, a point $u_* \in \mathcal{L}$ with $\operatorname{Im} u_* < 0$, and $\phi, s_0, \delta > 0$ such that*

- (P1) $F'(u_*) = 0$, i.e. $f'(u_*) = -ik_d\pi$; $f''(u_*) \neq 0$; $e^{2i\phi} f''(u_*) = -|f''(u_*)|$; $\sigma = \{u_* + se^{i\phi} : |s| \leq s_0\} \subset \mathcal{L} \cap \{\operatorname{Im} u < 0\}$;
- (P2) $H_{k_d}(u_* + se^{i\phi}) \leq H_{k_d}(u_*) - \frac{1}{4}|f''(u_*)|s^2$ for $|s| \leq s_0$;
- (P3) $H_{k_d}(u) \leq H_{k_d}(u_*) - \delta$ for $u \in \mathcal{L} \setminus \sigma$;
- (P4) $H_k(u) \leq H_{k_d}(u_*) - \delta$ for all $u \in \mathcal{L}$ and all $k \in \{1, 3, \dots, r-4\}$;
- (P5) $\mathcal{L}$ satisfies the hypotheses of Lemma 4.3 (i.e. $\inf_{\mathcal{L}} |u| > 0$, $\inf_{\mathcal{L}} \operatorname{Re} u > -\eta_1$, $\sup_{\mathcal{L}} |\arg(-u)| < \pi$).

Put $H_d = H_{k_d}(u_*) = \operatorname{Re} F(u_*)$, $\alpha = \operatorname{Im} F(u_*)$ and $B = e^{i\phi} G(u_*) \sqrt{2\pi/|f''(u_*)|} \neq 0$, $-iB = |B|e^{i\beta}$. Then

$$(4.10) \quad F_n = \frac{c_{r-2} \mathcal{A}_n |B|}{\pi} n^{1/2} e^{nH_d} \left[\cos(n\alpha + \beta) + o(1) \right] \quad (n \rightarrow \infty).$$

Consequently (a) $\limsup_{n \rightarrow \infty} \frac{1}{n} \log |F_n| \leq H_d$; (b) if $\alpha \notin \pi\mathbb{Z}$, then $F_n \neq 0$ for infinitely many n (indeed for at least one of any two consecutive sufficiently large n).

Proof. By Lemma 4.6 and Lemma 4.3 (applicable by (P5)), $I_{n,k} = n\mathcal{A}_n \int_{\mathcal{L}} e^{nF_k(u)} G(u) (1 + \varepsilon_n(u)) du$ for every k . For $k = k_d$, Lemma 4.5 applies with $F = F_{k_d}$ by (P1)–(P3) (its hypothesis (iv) follows from (P2), (P3) and the decay of $e^{F_k} G$ along the vertical rays, Lemma 4.4 applied with $n = 1$): $\int_{\mathcal{L}} = e^{nF(u_*)} [Bn^{-1/2} + o(n^{-1/2})]$. For $k \in \{1, 3, \dots, r-4\}$, (P4) gives $|e^{nF_k(u)}| \leq e^{(n-1)(H_d-\delta)} |e^{F_k(u)}|$ on $\mathcal{L}$, whence $\int_{\mathcal{L}} = O(e^{n(H_d-\delta)})$. Inserting into (4.8):

$$F_n = 2c_{r-2} \operatorname{Re} \left[\frac{n\mathcal{A}_n}{2\pi i} e^{nF(u_*)} (Bn^{-1/2} + o(n^{-1/2})) \right] + O(n\mathcal{A}_n e^{n(H_d-\delta)}),$$

which is (4.10) because $\operatorname{Re}[-ie^{in\alpha} B] = |B| \cos(n\alpha + \beta)$. (a) is immediate since $\log \mathcal{A}_n = O(\log n)$. For (b): if $|\cos x| < c$ and $|\cos(x + \alpha)| < c$ then, using $|\sin y| \geq \frac{2}{\pi} \operatorname{dist}(y, \pi\mathbb{Z})$, both $x - \frac{\pi}{2}$ and $x + \alpha - \frac{\pi}{2}$ lie within $\frac{\pi}{2}c$ of $\pi\mathbb{Z}$, so $\operatorname{dist}(\alpha, \pi\mathbb{Z}) \leq \pi c$. Taking $c = \frac{1}{2\pi} \operatorname{dist}(\alpha, \pi\mathbb{Z}) > 0$ we get $\max\{|\cos(n\alpha + \beta)|, |\cos((n+1)\alpha + \beta)|\} \geq c$ for every n , and (4.10) gives $F_n \neq 0$ for at least one of $n, n+1$ once n is large. $\square$

4.7. Two elementary bounds used in the verification.

Lemma 4.8 (segment bound). *Let $[a, b] \subset \overline{\Omega}$ be a segment meeting $\mathbb{R}$ at most in an endpoint lying in $(-\eta_1, 0)$, and let $d_\lambda = \operatorname{dist}([a, b], -\lambda)$ for $\lambda \in \{0, \eta_0\} \cup \{\eta_j, \eta_0 - \eta_j\}_{j \leq q}$ (all > 0). Put*

$B_2 = r(d_0^{-1} + d_{\eta_0}^{-1}) + \sum_j (d_{\eta_j}^{-1} + d_{\eta_0 - \eta_j}^{-1})$. For $N \geq 1$ let $u_\nu = a + \frac{\nu}{N}(b - a)$ and $h = |b - a|/N$. Then for every $k \geq 0$

$$\sup_{u \in [a, b]} H_k(u) \leq \max_{0 \leq \nu \leq N} H_k(u_\nu) + \frac{h}{2} \left(\max_{\nu} |f'(u_\nu)| + \frac{h}{2} B_2 + k\pi \right).$$

Proof. By (4.3), $|f''| \leq B_2$ on $[a, b]$, so $|f'(u) - f'(u_\nu)| \leq B_2|u - u_\nu|$ and $\sup |f'| \leq \max_{\nu} |f'(u_\nu)| + hB_2/2$. Along the unit-speed parametrisation, $|\frac{d}{ds} H_k| = |\operatorname{Re}(e^{i\theta} F'_k(u))| \leq |f'(u)| + k\pi$, and every point of the segment is within $h/2$ of a grid point. $\square$

Lemma 4.9 (tail bound). *Let $x_0 > -\eta_1$, $k \geq 0$, $Y \geq 1$ and $u = x_0 + iy$ with $|y| \geq Y$. Then*

$$H_k(u) \leq \beta_{\pm}(|y|) := -(r \pm k)\pi|y| + c_2 \log |y| + K, \quad (+ \text{ if } y > 0, - \text{ if } y < 0),$$

with $c_2 = r(|x_0| + x_0) + 2 \sum_j \eta_j - (q - r)\eta_0$ and $K = \kappa_0 + r(\frac{x_0^2}{Y} + |x_0|) + r(\frac{(\eta_0 + x_0)^2}{Y} + \eta_0 + x_0) + \sum_j (\frac{(\eta_j + x_0)^2}{Y} + \eta_j + x_0)$. If $c_2 \leq 0$ then $\beta_{\pm}$ is decreasing on $[Y, \infty)$, so $\sup_{|y| \geq Y} H_k(x_0 + iy) \leq \beta_{\pm}(Y)$.

Proof. For $a + x_0 > 0$: $\operatorname{Re}[(a + u) \log(a + u)] = (a + x_0) \log |a + u| - y \arg(a + u)$ with $\arg(a + u) = \operatorname{sgn}(y) \arctan \frac{|y|}{a + x_0}$, and $\arctan z \geq \frac{\pi}{2} - \frac{1}{z}$ for $z > 0$; hence $\operatorname{Re}[(a + u) \log(a + u)] \leq (a + x_0) \log(a + x_0 + |y|) - \frac{\pi}{2}|y| + (a + x_0)$. For the term $(-u) \log(-u)$: if $x_0 < 0$ the same computation with $a = 0$ applies; if $x_0 \geq 0$ then $-u = -x_0 + i(-y)$ has $|\arg(-u)| \geq \frac{\pi}{2}$ and $\operatorname{Re}(-u) \log |u| \leq 0$, giving $\operatorname{Re}[(a + u) \log(a + u)] \leq -\frac{\pi}{2}|y|$; in both cases $\operatorname{Re}[(a + u) \log(a + u)] \leq |x_0| \log(|x_0| + |y|) - \frac{\pi}{2}|y| + |x_0|$. For the terms with a minus sign ($b = \eta_0 - \eta_j$, $b + x_0 > 0$): $-\operatorname{Re}[(b + u) \log(b + u)] = -(b + x_0) \log |b + u| + y \arg(b + u) \leq -(b + x_0) \log |y| + \frac{\pi}{2}|y|$. Finally $\operatorname{Re}(ik\pi u) = -k\pi y$ and $\log(a + x_0 + |y|) \leq \log |y| + (a + x_0)/Y$. Summing: the terms $\mp \frac{\pi}{2}|y|$ cancel within each j -pair and add up to $-r\pi|y|$ from the two r -fold terms; the logarithmic coefficients add up to c_2 ; the constants to K . Monotonicity: $\beta'_{\pm}(y) = -(r \pm k)\pi + c_2/y < 0$ when $c_2 \leq 0$. $\square$

5. THE VERIFIED DATA AND THE PROOFS OF THEOREMS 1.1, 1.2, 1.3

5.1. The paths. For each of the three direction vectors we use a path $\mathcal{L}$ of the following shape. Let u_* be the saddle point of F_{k_d} in the lower half plane listed in Table 1, $e = e^{i\phi}$ with ϕ as in (P1), $s_0 = s_1 = 1$, and

$$\text{top} = u_* + s_0 e, \quad \text{bot} = u_* - s_0 e, \quad \text{ext} = u_* - (s_0 + s_1) e.$$

Then $\mathcal{L}$ is, in the order from $-i\infty$ to $+i\infty$: the vertical ray $\{\operatorname{Re} u = \operatorname{Re} \text{ext}, \operatorname{Im} u \leq \operatorname{Im} \text{ext}\}$; the segment $[\text{ext}, \text{bot}]$; the segment $\sigma = [\text{bot}, \text{top}]$ through u_* ; the segment $[\text{top}, x_0]$; the vertical ray $\{\operatorname{Re} u = x_0, \operatorname{Im} u \geq 0\}$. All points of $\mathcal{L}$ other than x_0 have $\operatorname{Im} u \neq 0$ (Table 2: $\operatorname{Im} \text{top} < 0$, and the segments $[\text{ext}, \text{bot}]$, $[\text{top}, x_0]$ meet $\mathbb{R}$ only at x_0). The constants of Lemma 4.3 are read off from Table 2: $\inf_{\mathcal{L}} |u| = \operatorname{dist}(\mathcal{L}, 0) > 1.7$, $\inf_{\mathcal{L}} \operatorname{Re} u = x_0 > -\eta_1$, $\sup_{\mathcal{L}} |\arg(-u)| < 2.02 < \pi$. Thus (P5) and the hypotheses of Lemma 4.6 hold.

TABLE 1. Saddle points u_k of $F_k = f + ik\pi u$ in the lower half plane ($f'(u_k) = -ik\pi$), the values $H_k(u_k)$, $f''(u_k)$, and $\alpha_k = \operatorname{Im} F_k(u_k) \bmod \pi$. The dominant mode is $k_d = r - 2$; $C_0 = -H_{k_d}(u_{k_d})$. Newton residuals $|f'(u_k) + ik\pi| < 10^{-49}$ at 50 digits.

	k	u_k	$H_k(u_k)$	$f''(u_k)$	$\alpha_k \bmod \pi$
(A)	1	$-6.538442182449 - 3.387946568365 i$	-980.968288736	$0.823837268 - 0.295203977 i$	0.205197
	3	$-0.590758658259 - 9.722688491285 i$	-939.030175174326	$0.219466732 - 0.473810380 i$	1.773388
(B)	1	$-4.485629507089 - 1.642867103557 i$	-1233.606836513	$1.791796334 - 0.483934337 i$	1.157548
	3	$-2.563290879243 - 4.779835583940 i$	-1213.240330004	$0.998222261 - 1.084898242 i$	2.977616
	5	$2.414446160185 - 6.698929974253 i$	-1175.784734481369	$0.003016599 - 0.866587262 i$	0.357148
(C)	1	$-3.204559550730 - 3.103130294133 i$	-230.897442276750	$0.748174102 - 0.426828129 i$	0.718053

TABLE 2. Path data and certified bounds. $\theta = H_d - \delta$ with $\delta = |f''(u_*)|s_0^2/8$ is the threshold in (P3), (P4). “sup” entries are the right-hand sides of Lemma 4.8 (grid points: $N = 20000$ on segments, $N = 400000$ on the rays up to $|\operatorname{Im} u| = Y$), “tail” entries are $\beta_{\pm}(Y)$ of Lemma 4.9 (with x_0 for the upper ray and $\operatorname{Re} \operatorname{ext}$ for the lower ray; $c_2 < 0$ in all cases).

	(A)	(B)	(C)
ϕ	2.139308100	2.354453992	1.830016624
$ f''(u_*) $	0.522170396	0.866592513	0.861363302
$M_3 = \sup_{ w \leq 1} f'''(u_* + w) \leq$	0.072404	0.202912	0.273693
$3 f''(u_*) /(2M_3) \ (\geq s_0 = 1)$	10.82	6.41	4.72
$\delta = f''(u_*) /8$	0.065271	0.108324	0.107670
$\theta = H_d - \delta$	-939.095446	-1175.893059	-231.005113
x_0	-2	-2	-3.460886528
top	-1.129137170 - 8.879985354i	1.708571167 - 5.990593547i	-3.460886528 - 2.136540163i
bot	-0.052380146 - 10.565391629i	3.120321153 - 7.407266402i	-2.948232574 - 4.069720425i
ext	0.485998366 - 11.408094766i	3.826196146 - 8.115602829i	-2.691905597 - 5.036310556i
Y	784	604	360
$\operatorname{dist}(\mathcal{L}, 0); \sup_{\mathcal{L}} \arg(-u)$	1.99; 1.614	1.70; 2.012	3.46; 1.564
sup H_{k_d} on $[\operatorname{top}, x_0]$	-939.296536	-1176.237722	-231.345825
sup H_{k_d} on upper ray, $0 \leq \operatorname{Im} u \leq Y$	-970.476007	-1227.711185	-235.559133
sup H_{k_d} on $[\operatorname{ext}, \operatorname{bot}]$	-939.281972	-1176.194938	-231.308192
sup H_{k_d} on lower ray, $-Y \leq \operatorname{Im} u \leq \operatorname{Im} \operatorname{ext}$	-939.979980	-1177.318472	-232.456263
tail $\beta_+(Y)$ (upper ray)	-18007.6	-21075.9	-4042.6
tail $\beta_-(Y)$ (lower ray)	-3119.3	-1443.7	-1768.0
sup H_1 on $\mathcal{L}$, $ \operatorname{Im} u \leq Y$; tails	-967.655; -13081.6, -8045.3	-1220.734; -13485.8, -9033.8	—
sup H_3 on $\mathcal{L}$, $ \operatorname{Im} u \leq Y$; tails	—	-1205.811; -17280.9, -5238.7	—

5.2. Verification of (P1)–(P5). (P1). u_* is the root of $f'(u) + ik_d\pi = 0$ in Table 1 (obtained from the exact integer polynomial $P(\tau) = (\tau - \eta_0)^r \prod_j (\tau - \eta_j) - \tau^r \prod_j (\tau - \eta_0 + \eta_j)$, whose roots are the solutions of $e^{f'(u)} = \pm 1$, $u = \tau - \eta_0$, followed by Newton iteration on $f'(u) + ik_d\pi$ itself; since $f''(u_*) \neq 0$ and the residual is $< 10^{-49}$, the Newton–Kantorovich theorem places an exact zero within 10^{-48} of the printed value, which does not affect any of the inequalities below). $f''(u_*) \neq 0$ by the table; $\phi = \frac{1}{2}(\pi - \arg f''(u_*))$ (plus π , so that $\operatorname{Im} e^{i\phi} > 0$), whence $e^{2i\phi} f''(u_*) = -|f''(u_*)|$. Since $s_0 = 1 < |\operatorname{Im} u_*|$, σ lies in the lower half plane.

(P2). For $|w| \leq s_0$ the disc $|u - u_*| \leq s_0$ lies in $\Omega \cap \{\operatorname{Im} u < 0\}$ and $|f'''| \leq M_3$ there, with M_3 bounded by (4.3) using the distances from u_* to the branch points minus s_0 (Table 2). Taylor’s formula gives $F(u_* + se^{i\phi}) = F(u_*) - \frac{1}{2}|f''(u_*)|s^2 + \rho(s)$, $|\rho(s)| \leq M_3|s|^3/6 \leq \frac{1}{4}|f''(u_*)|s^2$ for $|s| \leq s_0 \leq 3|f''(u_*)|/(2M_3)$, which holds (row 4 of Table 2).

(P3). On $[\operatorname{ext}, \operatorname{bot}]$, $[\operatorname{top}, x_0]$ and on the two rays truncated at $|\operatorname{Im} u| = Y$, Lemma 4.8 gives the certified suprema in Table 2, all $< \theta$. Beyond $|\operatorname{Im} u| = Y$, Lemma 4.9 applies (with x_0 , resp. $\operatorname{Re} \operatorname{ext}$, both $> -\eta_1$, and $c_2 < 0$ because $r(|x_0| + x_0) < (q - r)\eta_0 - 2 \sum_j \eta_j$, cf. (1.1)); the values $\beta_{\pm}(Y)$ are far below θ . Together with (P2) at the endpoints of σ (where $H_{k_d} \leq H_d - 2\delta$), this is (P3) with the stated δ .

(P4). The same checks, now for H_k with $k \in \{1, 3, \dots, r - 4\}$ and including the segment σ itself, give the last rows of Table 2, all $< \theta$.

(P5). See §5.1.

The phase. $\alpha = \operatorname{Im} F_{k_d}(u_*)$ has $\operatorname{dist}(\alpha, \pi\mathbb{Z}) \geq 0.357$ in all three cases (Table 1), so $\alpha \notin \pi\mathbb{Z}$.

5.3. Proof of Theorems 1.1, 1.2, 1.3. Fix one of the direction vectors (1.4)–(1.6). By §5.2, Proposition 4.7 applies, so with $C_0 := -H_d$ (Table 1, equation (1.7)) we have $\limsup \frac{1}{n} \log |F_n| \leq -C_0$ and $F_n \neq 0$ for infinitely many n . By Lemmas 3.2 and 3.3, $\Delta_n F_n \in \mathbb{Z} + \sum_s \mathbb{Z}\zeta(s)$ (s odd, $r + 2 \leq s \leq q - 2$) and $\log \Delta_n = n(C_2 + o(1))$ with C_2 as in §3.3.

Suppose, for contradiction, that $\zeta(r+2), \zeta(r+4), \dots, \zeta(q-2)$ are all rational, and let $A \in \mathbb{N}$ be a common denominator. Then $N_n := A \Delta_n F_n \in \mathbb{Z}$ for $n \geq n_0$, and

$$\log |N_n| \leq \log A + n(C_2 - C_0 + o(1)) \rightarrow -\infty$$

because $C_0 - C_2 > 0$ (equal to 2.1396, 4.5499, 3.7962 respectively, (1.7)). Hence $N_n = 0$, i.e. $F_n = 0$, for all large n , contradicting Proposition 4.7(b). Therefore at least one of $\zeta(r+2), \dots, \zeta(q-2)$ is irrational: for (A) one of $\zeta(7), \dots, \zeta(21)$; for (B) one of $\zeta(9), \dots, \zeta(33)$; for (C) one of $\zeta(5), \dots, \zeta(11)$. $\square$

Remark 5.1. (i) For (B) the dominant saddle has $\operatorname{Re} u_* = 2.41 > 0$, i.e. $\operatorname{Re} \tau_0 = \eta_0 + \operatorname{Re} u_* > \eta_0$ in Zudilin's variable; this is the situation excluded by the hypothesis “ $\operatorname{Re} \tau_0 < \eta_0$ ” of [11, Lemma 20] (stated there for $r = 3$ only). The path of §5.1 handles it because the exact integrand has no singularities in the open lower half plane, so the contour may leave the strip $-\eta_1 < \operatorname{Re} u < 0$ there. (ii) The r zero blocks and the $r-1$ derivatives are tied together: (P1)–(P3) concern the mode $k = r-2$, and for larger k the saddle moves away from the real axis ($\operatorname{Im} u_* = -3.1, -9.7, -6.7$ for $r = 3, 5, 7$), where the zeros of R_n no longer make $|\mathcal{G}|$ small; this is why the length of the list of zeta values grows with r . (iii) With Zudilin's own directions for $r = 3$ ($\eta_0 = 91$, $\boldsymbol{\eta}' = (27, 27, 27; 29, \dots, 38)$) the same programme returns $u_* = -3.520994582 - 3.328206906i$, $C_0 = 227.5801964127$, $C_2 = 226.2494426639$, reproducing [11, p. 285] to all published digits; the directions (1.6) merely give a larger margin.

6. WHAT IS VERIFIED BY COMPUTER, AND HOW

Every non-classical ingredient of the proof is one of finitely many explicit inequalities between computable real numbers, listed in Tables 1, 2 and in §3.3. They were checked with the Python programs `zud.py` and `certify.py` (available with this note) as follows.

- (1) *Saddle points and C_0 .* $P(\tau)$ is built with exact integer coefficients; its roots are computed with `mpmath.polyroots` at 90 digits; the roots with $\operatorname{Im} \tau > 0$, $\operatorname{Re} \tau > \eta_0 - \eta_{r+1}$ are mapped to $u = \tau - \eta_0 - i \operatorname{Im} \tau$ and polished by Newton's method on $f'(u) + ik\pi$ at 50 digits (residual $< 10^{-49}$); k is read off as $-\operatorname{Im} f'(u)/\pi$. $H_k(u_k)$, $f''(u_k)$, α_k and $G(u_k)$ are evaluated at 50 digits from (4.1), (4.3) with principal branches. Since $F'_k(u_k) = 0$, an error ϵ in u_k changes $H_k(u_k)$ by $O(\epsilon^2)$; all printed digits are reliable.
- (2) M_3 . From (4.3), $|f'''(u)| \leq r(|\eta_0 + u|^{-2} + |u|^{-2}) + \sum_j (|\eta_j + u|^{-2} + |\eta_0 - \eta_j + u|^{-2})$, maximised over the disc $|u - u_*| \leq s_0$ by replacing each distance by its minimum over the disc.
- (3) *Grid checks (Lemma 4.8).* H_k and $|f'|$ are evaluated in double precision at the $N+1$ grid points; B_2 is computed from the exact distances of the segment to the branch points; the reported “sup” is $\max_\nu H_k(u_\nu) + \frac{h}{2}(\max_\nu |f'(u_\nu)| + \frac{h}{2}B_2 + k\pi) + 10^{-9}$, the last term dominating the double-precision rounding error of the evaluations (the summands of H_k are of size $\leq 10^4$, so the rounding error is $< 10^{-10}$).
- (4) *Tails (Lemma 4.9).* c_2 , K and $\beta_\pm(Y)$ are elementary expressions in the data, evaluated in double precision; the slack is several hundred units.
- (5) *Arithmetic side.* The set X is generated exactly; the values φ_i are computed in exact rational arithmetic (and agree with an independent floating-point evaluation on every interval); $\psi(1+x_i)$ is computed to 30 digits; I_Φ is the finite sum (3.7). The total numerical uncertainty in C_2 is below 10^{-9} .

The margins $C_0 - C_2 \in \{2.14, 4.55, 3.80\}$ exceed the total numerical uncertainty by more than nine orders of magnitude; the inequalities (P2)–(P4) hold with slack at least 0.18 (dominant mode) and 28 (subdominant modes) on the finite parts and by thousands on the tails. The verification takes about one minute on a laptop.

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